

Flubenzole 5% w/w Premix Powder

Flubenzole 5% w/w Premix Powder appears to be a white to slightly yellow color powder.

Composition:

Each g contains

Flubendazole

50 mg

Pharmacodynamics:

Flubendazole is a synthetic anthelmintic belonging to the benzimidazole carbamates which acts by inhibiting the microtubular assembly in absorptive cells of nematodes. Flubendazole acts by binding to tubulin, the dimeric subunit protein of the microtubules. It inhibits microtubular assembly in absorptive cells: i.e. of intestinal cells of nematodes. This is shown by disappearance of cytoplasmic microtubules, accumulation of secretory granules in the cytoplasm due to a block in their transport, leading to an impaired coating of the cellular membrane and a decreased digestion and absorption of nutrients. Irreversible lytic degeneration of the cell, due to the accumulation of secretory substances (hydrolytic and proteolytic enzymes), results in the death of the parasite. These changes are relatively fast and are primarily seen in those organelles directly involved in the secretory and absorptive functions of the cells. In contrast the changes are not seen in host cells.

Pharmacokinetics:

Flubendazole is very poorly soluble in aqueous systems, such as the gastrointestinal tract, which results in a low dissolution rate and a very low absorption. This is reflected by the high faecal excretion of unchanged parent drug. The very small fraction absorbed is extensively metabolised by first pass metabolism in the liver, involving carbamate hydrolysis and ketone reduction. The biotransformation products are conjugated to glucuronides or sulphate conjugates and excreted in the bile and the urine. The excretion in urine is relatively low and consists almost exclusively of metabolites with only small amounts of unchanged compound. In pigs, highest tissue levels are measured in liver and kidneys. The half-life of flubendazole in tissues is 1 - 2 days.

Indications:

Pigs: Flubendazole is a broad spectrum anthelmintic for oral administration, active against mature and immature stages of the following nematodes of the gastrointestinal tract of the pig: *Ascaris suum* (large roundworm), *Hyostrogylus rubidus* (red stomach worm), *Oesophagostomum dentatum* (nodular worm), *Trichuris suis* (whip worm), *Strongyloides ransomi* (threadworm) (adult) and *Metastrongylus apri* (lungworm). Flubendazole is ovicidal.

Directions for Use:

For oral administration only.

For incorporation into dry feed at a registered mill.

A manufacturer who is approved to incorporate directly at any concentration, veterinary medicinal products or premixtures containing such products, must be responsible for mixing when incorporation is less than 2 kg per tonne for final feed.

Pigs:

Dosage

The standard recommended total dosage is 5 mg flubendazole per kg bodyweight. The amount of product to be incorporated should be calculated according to the average bodyweight of the pigs to be treated.

Incorporation and dosing instructions

If animals are to be treated collectively rather than individually, they should be grouped according to their bodyweight and dosed accordingly, in order to avoid under- or over-dosing. As a guide, the following incorporation rates are suggested:

Standard dosing regime

Incorporation:

Add 600 g of product to at least 5 kg of one of the feed ingredients and mix well. Thoroughly mix this premix with the remaining ingredients making in all one tonne of medicated feed, which can then be fed as mash or pellets. This gives 30 mg flubendazole per kg of finished feed.

Breeding stock should be treated for 10 consecutive days. Weaners and fattening pigs – should be treated for 5 consecutive days, or the event of a heavy *Trichuris* infestation, for 10 consecutive days.

Variable dosing regime

To facilitate feeding for different lengths of time to suit the intervals between feed deliveries, the standard dosage can be divided and administered over differing periods of time, as shown below.

Incorporation:

Add the required amount of product to at least 5kg of one of the feed ingredients. Thoroughly mix this premix with the remaining ingredients making in all one tonne of medicated feed, which can then be fed as mash or pellets.

a) Breeding stock

Amount of Flubenzole 5% w/w to add to each 5 kg premix for making up each tonne of final feed	Flubendazole inclusion rate in final feed (mg/kg)	Duration of treatment (days)	Total dose of flubendazole (mg/kg bodyweight)	Uses
400 g	20	14	5	<i>Ascaris suum</i> , <i>Oesophagostomum dentatum</i> and <i>Hyostrogylus rubidus</i>
300 g	15	21	5	
200 g	10	28	5	

b) Weaners and fattening pigs

Amount of Flubenzole 5% w/w to add to each 5 kg premix for making up each tonne of final feed	Flubendazole inclusion rate in final feed (mg/kg)	Duration of treatment (days)	Total dose of flubendazole (mg/kg bodyweight)	Uses
200 g	10	14	5	<i>Ascaris suum</i> , <i>Oesophagostomum dentatum</i> and <i>Hyostrogylus rubidus</i>
150 g	7.5	21	5	

In the event of a heavy *Trichuris* infestation, use 600 g/tonne (30 mg/kg final feed) for 10 days.

Treatment frequency:

Pigs should be treated twice a year unless recommended otherwise by a veterinary surgeon. Pigs brought onto the premises should be treated on arrival and before mixing with other animals.

Treatment of clinical worm infestations:

Treat relevant infestations at the following intervals:

Lungworm	every 3 - 4 weeks
Nodular worm	every 2 months
Large round worm	every 2 months
Red stomach worm	every month
Whipworm	every 6 weeks

Consult a veterinary surgeon for initial identification of problem species.

Mode of Administration:

Oral administration.

Contraindication:

None known.

Warning and Precautions:

Special warnings for target species

None.

Special precautions for use

None.

i. Special precautions for use in animals

None.

ii. Special precautions to be taken by the person administering the veterinary medicinal product to animals

Accidental ingestion by humans should be avoided. May cause sensitisation by skin contact. May cause skin and eye irritation. Avoid direct skin contact. Wear overalls, safety glasses and impervious gloves when mixing and handling the product. Wash affected parts if skin contact occurs. If accidental eye contact occurs, immediately rinse thoroughly with water. If the operations involve potential exposure to dust, wear either a disposable filter and half-mask respirator or a non-disposable respirator.

Interactions with Other Medicaments:

None known.

Statement on Usage During Pregnancy and Lactation:

Not applicable.

Adverse Effects / Undesirable Effects:

None known.

Overdose and Treatment:

Flubendazole is an analog of mebendazole for which the side effects of overdose include transient gastrointestinal abnormalities.

Withdrawal Period (s):**Pigs:**

Animals must not be slaughtered for human consumption during treatment. Pigs may be slaughtered for human consumption only after 7 days from the last treatment.

Storage Condition:

Store below 30°C. Protect from heat and sunlight exposure.

Keep medicine out of reach of children/Jauhkan ubat dari kanak-kanak.

Any unused veterinary medicinal product or waste materials derived from such veterinary medicinal products should be disposed of in accordance with local requirements.

Shelf-life as packaged for sale: 2 years

Shelf-life after opening: 7 days

Shelf-life after reconstitution: 24 hours

Packaging Available:

10 kg.

Date of revision:

02-09-2016

Name and Address of Manufacturer:

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